

Jiaying(Kary) Zheng

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EDUCATION

University of California, Irvine

Expected Jun 2027

Bachelor of Science in Computer Science; Bachelor of Arts in Psychological Science

Honors: ICS Honors Cumulative GPA: 3.95/4.0

PROFESSIONAL EXPERIENCE

Apache Texera, Open-Source Software Engineering Intern, UC Irvine, CA

May 2026–Present

- Implemented Python standalone-code generation for 148 Texera workflow operators, enabling execution independent of the native Scala/JVM engine.
- Designed an integration-based verification framework comparing native Texera execution against generated Python outputs in GitHub CI.
- Optimized workflow verification with persistent worker pools, batching, and parallel execution, reducing runtime from 32.8s to 8.0s (~4×) in a benchmark of the first 53 implemented operators.
- Collaborate closely with Prof. Chen Li and other contributors on system design, technical handoffs, GitHub issues, pull requests, and production-quality open-source code.

Cognitive Anteater Robotics Laboratory, Research Assistant, UC Irvine, CA

Sep 2024–June 2026

- Designed Q-learning simulations across 4 experimental conditions, achieving stable convergence under 130-second delays.
- Improved decision accuracy by ~20–30% through reward tuning, punishment modeling, and cross-environment evaluation.
- Co-authored two peer-reviewed publications on episodic-like memory and delayed gratification in simulated cuttlefish behavior.

Mucci Assessment, Intern, Remote

Sep 2024–Sep 2025

- Integrated Google Sheets API to automatically sync website content, replacing manual content updates.
- Built reusable web modules for client projects, reducing duplicate development across similar website features.

Learning Assistant, UC Irvine, CA

Apr 2025–June 2025

- Led weekly Python labs for 30+ students, reinforcing Python programming and core CS concepts.
- Guided students through debugging, program design, and problem-solving during hands-on lab sessions.

PROJECTS

Scalable Search Engine, Irvine, CA

Jan 2026–Mar 2026

- Built a search engine that indexed 50K+ web pages via a custom web crawler, achieving sub-300ms query latency with optimized retrieval.
- Designed a disk-based inverted index with partial indexing and k-way merging, reducing memory usage at scale.
- Developed TF-IDF ranking with HTML tag weighting and term proximity, improving relevance for ranked results.

Fabflix (Full-Stack Web Application), Irvine, CA

Jan 2026–Mar 2026

- Built a full-stack movie platform with search, authentication, checkout, and shopping cart features.
- Developed RESTful APIs using Java Servlets, Tomcat, and MySQL to support modular backend services.
- Optimized MySQL queries and full-text search, reducing query response time by ~40%.
- Deployed Dockerized microservices on Kubernetes and AWS, enabling load-balanced backend performance.

LifeLink, AI-Powered CPR Emergency Platform, Los Angeles, CA

Apr 2026

- Built an AI CPR voice agent with ElevenLabs, delivering real-time CPR instructions through a React/TypeScript frontend.
- Connected SOS activation, AED routing, volunteer alerts, and patient handoff into a coordinated emergency workflow.

PUBLICATIONS

Kandimalla, S., Wong, Q. Y., **Zheng, K.** et al. Episodic-like memory in a simulation of cuttlefish behavior. *Scientific Reports*, 2025.

Wong, Q. Y., **Zheng, K.** et al. "Delayed Gratification in a Robotic Model of Cuttlefish Behavior." *Lecture Notes in Artificial Intelligence*, Springer Nature, 2026.

SKILLS

Languages & Frameworks: Python, Java, Scala, C++, JavaScript/TypeScript, SQL, React, Flask, Pandas, Java Servlets.

Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Docker, Kubernetes, AWS

Tools & Concepts: Git, GitHub Actions, CI/CD, REST APIs, Information Retrieval, Machine Learning, Data Structures & Algorithms